WIRELESS SYSTEMS LAB

EXPERIMENT 10 · CONCEPT

01 / 03

One spectrum. Many users.

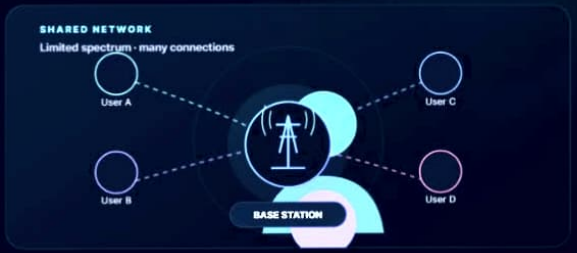
How mobile networks coordinate simultaneous access to a shared wireless medium.

THE CORE IDEA

Multiple access techniques divide or distinguish limited radio resources so many users can communicate through the same network—efficiently, reliably and with controlled interference.

- **Frequency** — Divide spectrum
- **Time** — Schedule access
- **Code** — Separate signals

Why it matters Capacity · fairness · spectral efficiency · quality of service



01 **FDMA**
FREQUENCY DIVISION
Separate frequency bands

02 **TDMA**
TIME DIVISION
Separate time slots

03 **CDMA**
CODE DIVISION
Unique spreading codes

04 **OFDMA**
SUBCARRIERS DIVISION
Orthogonal carrier groups

MULTIPLE ACCESS TECHNIQUES

WIRELESS SYSTEMS LAB

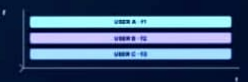
EXPERIMENT 10 · WORKING PRINCIPLES

Four ways to share the air.

Frequency-versus-time views reveal how each method separates simultaneous users.

FDMA

FREQUENCY SLICES



Each user receives a dedicated frequency channel for the entire call.

- Guard bands reduce adjacent-channel interference.

TDMA

TIME SLICES



Users take turns on one carrier in repeating, synchronized time slots.

- Guard time prevents slot overlap.

CDMA

CODE SEPARATION



All users overlap in time and frequency but use distinct spreading codes.

- Correlation recovers the intended signal.

OFDMA

SUBCARRIER GROUPS



A scheduler assigns orthogonal subcarriers to users in time–frequency blocks.

- Orthogonality limits inter-carrier interference.

User A · User B · User C

Axes are conceptual · allocations not to scale

MULTIPLE ACCESS TECHNIQUES

From dedicated channels to dynamic grids.

The evolution of multiple access tracks the demand for capacity, flexibility and spectral efficiency.

1G FDMA — Analog voice

2G TDMA — GSM

2G / 3G CDMA — IS-95 · WCDMA

4G / 5G OFDMA — LTE · NR downlink

COMPARISON POINT	FDMA	TDMA	CDMA	OFDMA
Resource split	Frequency bands	Time slots	Spreading codes	Orthogonal subcarriers
Time allocation	Continuous	Slotted	Simultaneous	Dynamic time blocks
Bandwidth use	Fixed / lower	Shared / moderate	Wideband / high reuse	Flexible / very high
Interference	Adjacent-channel	Slot overlap	Multi-user / near–far	ICI if sync is lost
Key advantage	Simple; low delay	Efficient scheduling	Robust; soft capacity	High efficiency; scalable
Main limitation	Guard-band waste	Strict synchronization	Complex power control	High PAPR; sync needs
Typical use	**1G analog**	**GSM / 2G**	**IS-95, cdma2000, WCDMA**	**4G LTE, 5G NR DL**

KEY TAKEAWAY

Multiple access is the coordination layer that turns scarce spectrum into a shared, scalable service.

MORE USERS • BETTER EFFICIENCY • ADAPTIVE CAPACITY